

Examples Illustrating The Importance Of Effective Communication

Effective communication examples can be stated as active listening, giving and taking feedback, empathy, and respectfulness, responding to messages, having volume and clarity in messages, understanding non-verbal data, building friendliness and confidence, adapting your communication style to the audience, and so on.

Here are some examples of effective communication:

- Active listening
- Adapting your communication style to the listeners
- Friendliness
- Giving and receiving constructive feedback
- Confidence
- Volume and clarity
- Empathy and respect
- Understanding non-verbal messages
- Responsiveness

Dimensions of Communication

The various directions or dimensions of communication are as follows:

i. Downward Communication:

Downward communication means flow of information from higher levels to lower levels in the organizational hierarchy. Information flows from top-most authority to the bottommost persons (workers) through various levels. This flow of information generally prevails where autocratic style of management is dominant.

Contents of Information:

The following information generally flows in downward direction:

- How to do a job.
- Rationale for doing those jobs.
- Policies and practices of organisation.
- Performance level of employees.
- The need to develop a sense of mission.

Merits

- It provides information regarding organizational plans and policies.

- It informs employees about the rationale of organizational goals and how they can coordinate these goals with personal goals.
- It enables employees to know their area of discretion. They know what is expected of them and perform within those constraints.
- It increases job satisfaction and morale of employees to perform better.

ii Upward Communication:

Flow of information from lower levels to higher levels is known as upward communication. Employees respond to managers' directions and instructions through upward communication. This flow of communication is suitable where democratic or participative style of management is prevalent.

Contents of Information:

The following information generally flows in upward direction:

- Reports about subordinates' work, achievements, progress and future plans.
- Subordinates' work problems which need the assistance of supervisors.
- Ideas or suggestions offered by subordinates to improve the working of any unit or organisation as a whole.

- Subordinates' perception about their work, work environment and co-workers.
- Subordinates' perception about their work, work environment and co-workers.
- **Merits**
- Managers receive feedback from lower levels that helps in carrying out the controlling function.
- It helps in knowing attitude of employees towards corporate goals and plans for effective implementation of plans.
- It provides managers with creative ideas and suggestions from subordinates.
- It strengthens superior-subordinate relationships and creates harmonious industrial relations.
- It overcomes resistance to change on the part of employees as they can present their fears and apprehensions upwards.

iii Horizontal or Lateral Communication:

Flow of information amongst people at the same level is known as horizontal communication. It is interaction amongst peer groups. Most common horizontal communication is communication between functional heads. Marketing manager and production manager constantly interact with each other to coordinate demand with production schedules.

It takes place between:

- i. Members of a particular work group.
- ii. Members of different work groups.

- iii. Members of different departments operating at the same level.
- iv. Members of line and staff.
- v. It does not follow the official chain of command but cuts across it.

Merits

- It helps to review activities assigned to people working at identical positions.
- It speeds up the flow of work.
- It facilitates problem-solving amongst members at their level and integrates the work.
- It develops trust and confidence amongst people of different departments.
- It provides job satisfaction to employees and boosts their morale to improve performance.

iv Diagonal or Cross-Wise Communication:

Flow of information amongst people of different departments at different levels is known as diagonal communication. When regional sales manager talks to workers of finance or production department, diagonal communication is said to have taken place. People who are generally not in contact, come close to each other through diagonal communication.

Forms of Diagonal Communication:

Diagonal communication can be oral and written. Oral communication takes place in meetings, conferences, project teams etc. Written communication takes place through magazines, bulletins, boards, notices, circulars etc.

Merits

- It increases efficiency and speed of work. People can directly talk to each other without following the formal chain of command.
- It boosts morale and commitment of employees to work as they can directly talk to the managers.
- It coordinates the activities of various departments as they can interact with each other.
- It provides job satisfaction to employees.

Stages of Communication

Sender – Sender or Communicator is the person who sends a message or idea. He is the source and initiates the process of communication. Sender may be a speaker, a writer or an actor.

Message – Message is what is conveyed by the sender. It consists of words, facts, ideas and opinions, etc. it is the subject – Matter of communication.

Encoding – It is the use of appropriate verbal or non-verbal language for transmitting the message.

In order to transmit the idea, the sender translates the idea into a language (words, symbols, gestures) known to both the parties.

Channel – It is the medium or route through which the message is passed from the sender to the receiver. It may be face-to-face talk, telephone, letter, radio, television, etc.

Receiver – Receiver or communicatee is the person or group who is supposed to receive the message. He may be a listener, a reader or an observer.

Decoding – It means translating the message into words for the purpose of understanding. The receiver interprets the message to derive its meaning.

Feedback – It refers to reaction, reply, response, which the receiver sends to acknowledge his understanding of the message. It may consist of words, actions or facial expressions (e.g., nodding the head). When the sender receives the feedback, the process of communication is said to be complete.

Oral communication

Oral communication is transfer of information from sender to receiver by means of verbal and visual aid. Examples of oral communication include presentations, speeches, discussions, etc.

Advantages

- Immediate feedback
- Time saving
- Economical

- Personal touch
- Flexibility
- Secrecy
- Group communication
- Effectiveness
- Persuasive power

Verbal communication

The communication in which the sender uses words, whether spoken or written, to transmit the message to the receiver is known as Verbal Communication. It is the most effective form of communication that leads to the rapid interchange of information and feedback. There are fewer chances of misunderstanding as the communication between parties is clear, i.e. the parties are using words for saying anything.

The communication can be done in two ways (i) Oral – like face to face communication, lectures, phone calls, seminars, etc. (ii) Written – Letters, E- mail, SMS, etc. There are two types of communication, they are:

- **Formal Communication:** Also termed as official communication, it is a type of communication in which the sender follows a pre-defined channel to transmit the information to the receiver is known as formal communication.

- **Informal Communication:** Most commonly known as grapevine, the type of communication in which the sender does not follow any pre-defined channels to transmit the information is known as informal communication.

Non-verbal communication

Non-verbal communication is based on the understanding of the parties to communication, as the transmission of messages from the sender to receiver is wordless i.e. the communication uses signs. So, if the receiver understands the message completely and proper feedback is given afterwards, then the communication succeeds.

It complements the verbal communication many times, to understand the mindset and the status of the of the parties, which is not spoken by them, but it is an act of understanding. The types of Non-verbal communication are as under:

Chronemics: The use of time in communication is chronemics, which speaks about the personality of the sender / receiver like punctuality, the speed of speech, etc.

Vocalics: The volume, tone of voice and pitch used by the sender for communicating a message to the receiver is known as vocalics or paralanguage.

Haptics: The use of touch in communication is the expression of feelings and emotions.

Kinesics: It is the study of the body language of a person, i.e., gestures, postures, facial expressions, etc.

Proxemics: The distance maintained by a person while communicating with others, communicates about the relationship of the person with others like intimate, personal, social and public.

Artifacts: The appearance of a person speaks about his personality, i.e. by way of clothing, carrying jewellery, lifestyle, etc. This kind of communication is known as artifactual communication.

casual communication

A casual style of communication, as the term implies, brings to mind informal conversation, often between friends and acquaintances. It is easy to approach, free of all pretense, and is not used for high-pressure conversations. You would want to use a casual communication style with people you are familiar with and trust.

Casual communication occurs with friends, family, and individuals in business, entertainment, and other walks of life. It is commonly used in social events such as a friend's birthday, a wedding, or a party. The casual communication style tends to be informal and relaxed, but not "rude" or without manners. It uses slang languages or expressions that can easily be understood without giving too much information.

In casual communication, the speaker is comfortable and at ease. The listener is focused on the speaker's message rather than how they express it. The casual style of speaking is also characterized as informal, candid, and timely and does not often address the conventional communication rules.

You can use the casual speech style in the following scenarios.

- When speaking with family and friends
- When the audience is well-known to you

- When speaking to your employees and client
- When speaking to your business partners

Formal communication

Formal communication is, typically, conveyed from the top leadership to various departments and employees. Usually, every organization follows a procedure for formal conversation. Think about the annual meetings or even team meetings that your manager calls for. These are examples of formal communication.

Formal Communication Often Follows A Specific Structure Or Channels Such As Emails To the Clients. Formal Communication Is Time-Consuming. Formal Communication Is More Reliable, As It Follows A Pattern Set By The Organization. Examples of formal communication are - business letters, reports, orders, etc.

Types Of Formal Communication:

Vertical

Here, the communication is held between different organizational levels. So the message is either transferred from the juniors to the team leads to the manager or vice-versa.

Horizontal Or Lateral

This is the communication that happens between peers from different departments.

Crosswise Or Diagonal

As the name suggests, here the conversation takes place between two employees working at different levels in different departments. For example, a website developer discussing a project with a sales manager can be categorized as crosswise or diagonal communication.

Informal communication

Informal communication can also play a much larger role than just generating friendly chatter. This form of communication can be very useful in resolving a conflict between the employees and the management. Informal Communication Can Often Flow Freely In Any Direction.

Informal Communication Is Usually Quick And Easy To Navigate. Informal Communication Takes Off On Its Own And Sets Its Own Course. Examples of informal communication are face-to-face communication, telephonic conversations, etc.

Types Of Informal Communication:

Single Strand Chain

This is the type of communication where A shares an idea or information with B, who then passes it to C, and so on.

Cluster Chain

Have you ever noticed how a social media challenge becomes viral? People start something unique and tag, say, three friends for the challenge. They complete the challenge and tag three more people each, and so on. That's how a cluster chain communication is formed and continues.

Gossip Chain

Think of the college canteen conversations, where one person vividly describes her recent adventures to a group of friends gathered around the table to listen. That's how the gossip chain works. One person initiates the conversation and shares information with a group of people, who then pass on the information to more people.

Interactive communication

The interactive model of communication refers to the actual process of communication as messages are sent back and forth between the sender and receiver. The cyclical process includes a source that sends the message to the receiver and a reaction from the receiver in the form of feedback, which is the message that is then received by the original sender. A simple conversation over the phone is the most basic example of the interactive model. It consists of two participants that are both consistently sending and receiving messages based on reactions from previous messages. This interaction process also considers the fields of experience of both participants, which include the knowledge and values of each of the participants and may impact the meaning of the mess. The interactive model is more complex than the linear model, as it considers the feedback given from the receiver of the message. **Feedback** refers to the reaction to the message which may be verbal or nonverbal.

Features of communication, which are the sender, message, receiver, and feedback. It also includes more complex features, such as fields of expertise, physical context, psychological context, and channels.

An example of the interactive model of communication is the conversation that takes place between employees during a training session. As the trainee asks questions in regard to the job, the other worker replies with an answer. A worker in training is likely to continuously ask questions about the job, prompting responses from the other worker.

Listening

Listening is the ability to accurately receive and interpret messages in the communication process.

Listening is key to all effective communication, without the ability to listen effectively messages are easily misunderstood. Listening is one of the most important skills you can have. An active process of getting information, ideas. “Listening is the process of receiving, constructing meaning from, and responding to spoken.

Process Of Listening



Receiving

- Is the intentional focus on hearing a speaker’s message.
- This stage is represented by the ear because it is the primary tool involved with this stage of the listening process.

Understanding

- In the understanding stage, we attempt to learn the meaning of the message, which is not always easy.
- Deciding what the message means to you.

Remembering

- Remembering begins with listening; if you can’t remember something that was said, you might not have been listening effectively.

Evaluating

- The fourth stage in the listening process is evaluating.
- evaluations of the same message can vary widely from one listener to another.

Responding

- Responding—sometimes referred to as feedback—is the fifth and final stage of the listening process.
- Your reaction to the message. It can be emotional and intellectual.

Why Listening is Important?

- To avoid communication errors.
- Helps to learn something new.
- It is the key to success.

Five major forms of communication:

Intrapersonal Communication

Dyadic Communication

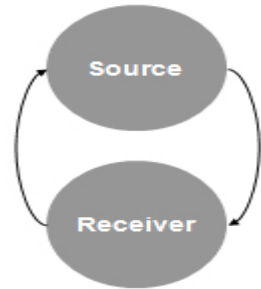
Small Group Communication

Public Communication

Mass Communication

Intrapersonal Communication

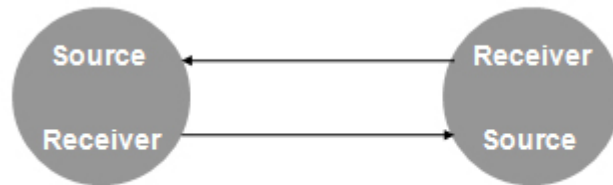
Intrapersonal communication is a communication which happens yourself. Here both Source (sender) and receiver is only one. so, the feedback works without any interruption. Example: A person can communicate himself through pain, thinking, feelings and emotion etc.



A. Intrapersonal communication

Dyadic Communication

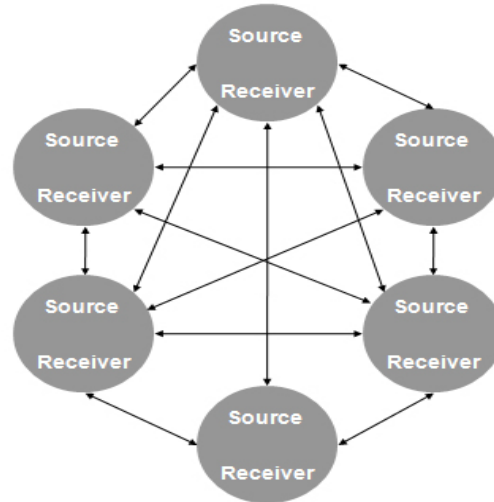
In Dyadic communication, two persons are involved in this communication process. Here the Source becomes a receiver and receiver become Source because of dynamic communication process where the feedback's are shared between Source and the receiver.



B. Dyadic communication

Small Group Communication

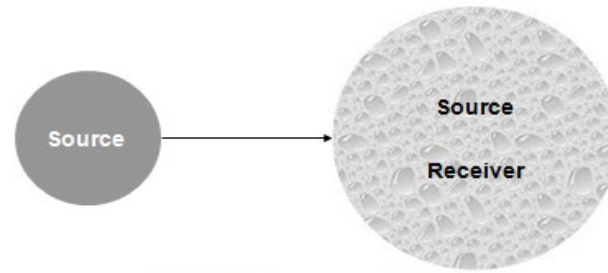
More than two members involved in communication process will become a group communication. If least number of persons is involved in the group communication is called as small group communication. In this communication process, everyone becomes a Source as well as receiver through sharing information and gives feedback to another.



C. Small group communication

Public Communication

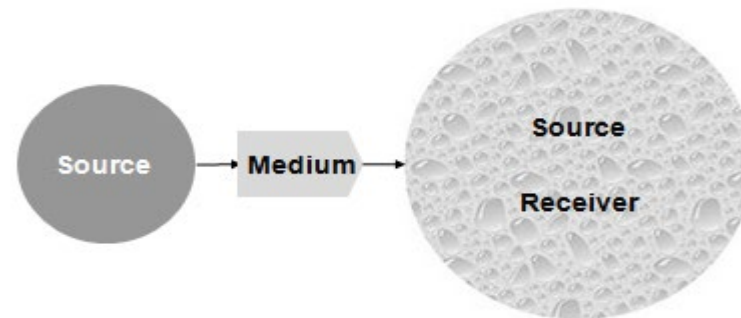
In public communication, Source or messages from a single person will reach or received by huge number of audience. But in this communication there is no mutual feedbacks between source and receiver like small group communication and it's only focused on Speaker.



D. Public communication

Mass Communication

In mass communication, basically have a large number of audience and they are all can't grouped together in one place so we need certain tool or technology for communication process. But in mass communication, there is no direct access with receiver. So, for that they need media like newspaper, radio, television and internet. Here the audience feedback is very less or delayed.



E. Mass communication

Various contexts for speaking

Conference

- A conference is a gathering of people with a common interest or background, with the purpose of allowing them to meet one another and learn about and discuss issues, ideas, and work on a topic of mutual concern.
- Conference may be held in place other than workplaces to avoid distractions
- Can be held in other area of country or the world
- Conference may be held in cyberspace, or something similar
- These presentation may take place in the form of lectures, slide show or films, workshop, panel discussion or interactive discussion
- A conference may lasts for few hours or several days
- May be onetime event or regular(Annual) fixture on participants' schedules

Seminar

Seminar may be defined as gathering for the purpose of discussing a stated topic.

Such gatherings is usually interactive session, where the participants engage in discussion about the topic.

The discussion are headed by 1 or 2 presenters.

PURPOSE

- A seminar may have several purposes or just one
- Seminar may be for the purpose of education such as a lecture, where participants engage in discussion of an academics ,with the aim of gaining better insight on subject.
- May be held to impart skills r knowledge to participants
- Seminars can be motivational where purpose is to inspire the attendees to become better people.
- Seminars are for businessmen and women or likeminded people to network and meet other attendees with similar interests
- A trade seminar brings a wide cross section of the community together,as government officials, businessmen and general public.

Guidelines for preparation of good presentation slides

- Use the slide master feature to create a consistent and simple design template. It is fine to vary the content presentation (bulleted list, two-column text, text and image, etc.), but be consistent with other elements such as font, colors and background.
- Simplify and limit the number of words on each screen. Use key phrases and include only essential information.
- Limit punctuation and avoid putting words in all-capital letters. Empty space on the slide will enhance readability.
- Use contrasting colors for text and background. Light text on a dark background is best. Patterned backgrounds can reduce readability.
- Avoid the use of flashy transitions such as text fly-ins. These features may seem impressive at first but are distracting and get old quickly.
- Overuse of special effects such as animation and sounds may make your presentation “cutesy” and could negatively affect your credibility.
- Use good-quality images that reinforce and complement your message. Ensure that your image maintains its impact and resolution when projected on a larger screen.

- Limit the number of slides.
- Make sure the slides are readable from the back row seats. Text and graphic images should be large enough to read but not so large as to appear “loud.

Fonts

- Select a single sans-serif fonts such as Arial or Helvetica. Avoid serif fonts such as Times New Roman or Palatino because these fonts are sometimes more difficult to read.
- Use no font size smaller than 24 point.
- Use the same font for all your headlines.
- Select a font for body copy and another for headlines.
- Use bold and different sizes of those fonts for captions and subheadings.
- Add a fourth font for page numbers or as a secondary body font for sidebars.
- Don't use more than four fonts in any one publication.

- Clearly label each screen. Use a larger font (35-45 points) or different color for the title.
- Use larger fonts to indicate importance.
- Use different colors, sizes and styles (e.g., bold) for impact.
- Avoid italicized fonts as these are difficult to read quickly.
- Avoid long sentences.
- Avoid abbreviations and acronyms.
- Limit punctuation marks.
- No more than 6-8 words per line
- For bullet points, use the 6 x 6 Rule. One thought per line with no more than 6 words per line and no more than 6 lines per slide
- Use dark text on light background or light text on dark background. However, dark backgrounds sometimes make it difficult for some people to read the text.
- Do not use all caps except for titles.

- Put repeating elements (like page numbers) in the same location on each page of a multi-page document.
- To test the font, stand six feet from the monitor and see if you can read the slide

Design and Graphical Images

- Use design templates.
- Standardize position, colors, and styles.
- Include only necessary information.
- Limit the information to essentials.
- Content should be self-evident
- Use colors that contrast and compliment.
- Too many slides can lose your audience.
- Keep the background consistent and subtle.
- Limit the number of transitions used. It is often better to use only one so the audience knows what to expect.

- Use a single style of dingbat for bullets throughout the page.
- Use the same graphical rule at the top of all pages in a multi-page document.
- Use one or two large images rather than several small images.
- Prioritize images instead of a barrage of images for competing attention.
- Make images all the same size.
- Use the same border.
- Arrange images vertically or horizontally.
- Use only enough text when using charts or graphical images to explain the chart or graph and clearly label the image.
- Keep the design clean and uncluttered. Leave empty space around the text and graphical images.
- Use quality clipart and use it sparingly. A graphical image should relate to and enhance the topic of the slide.

- Try to use the same style graphical image throughout the presentation (e.g., cartoon, photographs)
- Limit the number of graphical images on each slide.
- Repetition of an image reinforces the message. Tie the number of copies of an image to the numbers in your text.
- Resize, recolor, reverse to turn one image into many. Use duplicates of varying sizes, colors, and orientations to multiply the usefulness of a single clip art image.
- Make a single image stand out with dramatic contrast. Use color to make a dramatic change to a single copy of your clip art.
- Check all images on a projection screen before the actual presentation.
- Avoid flashy images and noisy animation effects unless it relates directly to the slide.

Color

- Limit the number of colors on a single screen.
- Bright colors make small objects and thin lines stand out. However, some vibrant colors are difficult to read when projected.
- Use no more than four colors on one chart.
- Check all colors on a projection screen before the actual presentation. Colors may project differently than what appears on the monitor.

Written communication

- Research report is considered a major component of the research study for the research task remains incomplete till the report has been presented and/or written.
- As a matter of fact even the most brilliant hypothesis, highly well designed and conducted research study, and the most striking generalizations and findings are of little value unless they are effectively communicated to others.
- The purpose of research is not well served unless the findings are made known to others.
- Research results must invariably enter the general store of knowledge.
- Writing of report is the last step in a research study and requires a set of skills somewhat different from those called for in respect of the earlier stages of research.
- This task should be accomplished by the researcher with utmost care; he may seek the assistance and guidance of experts for the purpose.

MECHANICS OF WRITING A RESEARCH REPORT

There are very definite and set rules which should be followed in the actual preparation of the research report or paper. Once the techniques are finally decided, they should be scrupulously adhered to, and no deviation permitted. The criteria of format should be decided as soon as the materials for the research paper have been assembled. The following points deserve mention so far as the mechanics of writing a report are concerned:

1. **Size and physical design:** The manuscript should be written on unruled paper $8\frac{1}{2} \times 11$ in size. If it is to be written by hand, then black or blue-black ink should be used. A margin of at least one and one-half inches should be allowed at the left hand and of at least half an inch at the right hand of the paper. There should also be one-inch margins, top and bottom. The paper should be neat and legible. If the manuscript is to be typed, then all typing should be double-spaced on one side of the page only except for the insertion of the long quotations.
2. **Procedure:** Various steps in writing the report should be strictly adhered

3. **Layout:** Keeping in view the objective and nature of the problem, the layout of the report should be thought of and decided and accordingly adopted
4. **Treatment of quotations:** Quotations should be placed in quotation marks and double spaced, forming an immediate part of the text. But if a quotation is of a considerable length (more than four or five type written lines) then it should be single-spaced and indented at least half an inch to the right of the normal text margin.
5. **The footnotes: Regarding footnotes one should keep in view the followings:**
 - (a) The footnotes serve two purposes viz., the identification of materials used in quotations in the report and the notice of materials not immediately necessary to the body of the research text but still of supplemental value. In other words, footnotes are meant for cross references, citation of authorities and sources, acknowledgement and elucidation or explanation of a point of view. It should always be kept in view that footnote is not an end nor a means of the display of scholarship. The modern tendency is to make the minimum use of footnotes for scholarship does not need to be displayed.

(b) Footnotes are placed at the bottom of the page on which the reference or quotation which they identify or supplement ends. Footnotes are customarily separated from the textual material by a space of half an inch and a line about one and a half inches long.

(c) Footnotes should be numbered consecutively, usually beginning with 1 in each chapter separately. The number should be put slightly above the line, say at the end of a quotation. At the foot of the page, again, the footnote number should be indented and typed a little above the line. Thus, consecutive numbers must be used to correlate the reference in the text with its corresponding note at the bottom of the page, except in case of statistical tables and other numerical material, where symbols such as the asterisk (*) or the like one may be used to prevent confusion.

(d) Footnotes are always typed in single space though they are divided from one another by double space.

6. Documentation style: Regarding documentation, the first footnote reference to any given work should be complete in its documentation, giving all the essential facts about the edition used. Such documentary footnotes follow a general sequence. The common order may be described as under:

(i) *Regarding the single-volume reference*

1. Author's name in normal order (and not beginning with the last name as in a bibliography) followed by a comma;
2. Title of work, underlined to indicate italics;
3. Place and date of publication;
4. Pagination references (The page number).

Example

John Gassner, *Masters of the Drama*, New York: Dover Publications, Inc. 1954, p. 315.

(ii) *Regarding multivolumed reference*

1. Author's name in the normal order;
2. Title of work, underlined to indicate italics;
3. Place and date of publication;
4. Number of volume;
5. Pagination references (The page number).

(iii) *Regarding works arranged alphabetically*

- For works arranged alphabetically such as encyclopedias and dictionaries, no pagination
- reference is usually needed. In such cases the order is illustrated as under:

Example 1

- “Salamanca,” *Encyclopaedia Britannica*, 14th Edition.

Example 2

- “Mary Wollstonecraft Godwin,” *Dictionary of national biography*.
- But if there should be a detailed reference to a long encyclopedia article, volume and
- pagination reference may be found necessary.

(iv) *Regarding periodicals reference*

1. Name of the author in normal order;
2. Title of article, in quotation marks;
3. Name of periodical, underlined to indicate italics;
4. Volume number;
5. Date of issuance;
6. Pagination.

(v) *Regarding anthologies and collections reference*

Quotations from anthologies or collections of literary works must be acknowledged not only by author, but also by the name of the collector.

(vi) *Regarding second-hand quotations reference*

In such cases the documentation should be handled as follows:

1. Original author and title;
2. “quoted or cited in,”;
3. Second author and work.

Example

- J.F. Jones, *Life in Ploynesia*, p. 16, quoted in *History of the Pacific Ocean area*, by R.B. Abel, p. 191.

(vii) *Case of multiple authorship*

If there are more than two authors or editors, then in the documentation the name of only the first is given and the multiple authorship is indicated by “et al.” or “and others”.

Subsequent references to the same work need not be so detailed as stated above. If the work is cited again without any other work intervening, it may be indicated as *ibid*, followed by a comma and the page number. A single page should be referred to as p., but more than one page be referred to as pp. If there are several pages referred to at a stretch, the practice is to use often the page number, for example, pp. 190ff, which means page number 190 and the following pages; but only for page 190 and the following page ‘190f’. Roman numerical is generally used to indicate the number of the volume of a book. Op. cit. (opera citato, in the work cited) or Loc. cit. (loco citato, in the place cited) are two of the very convenient abbreviations used in the footnotes. Op. cit. or Loc. cit. after the writer’s name would suggest that the reference is to work by the writer which has been cited in detail in an earlier footnote but intervened by some other references.

7. Punctuation and abbreviations in footnotes:

The first item after the number in the footnote is the author's name, given in the normal signature order. This is followed by a comma. After the comma, the title of the book is given: the article (such as "A", "An", "The" etc.) is omitted and only the first word and proper nouns and adjectives are capitalized. The title is followed by a comma. Information concerning the edition is given next. This entry is followed by a comma. The place of publication is then stated; it may be mentioned in an abbreviated form, if the place happens to be a famous one such as Lond. for London, N.Y. for New York, N.D. for New Delhi and so on. This entry is followed by a comma. Then the name of the publisher is mentioned and this entry is closed by a comma. It is followed by the date of publication if the date is given on the title page. If the date appears in the copyright notice on the reverse side of the title page or elsewhere in the volume, the comma should be omitted and the date enclosed in square brackets [c 1978], [1978]. The entry is followed by a comma. Then follow the volume and page references and are separated by a comma if both are given. A period closes the complete documentary reference. But one should remember that the documentation regarding acknowledgements from magazine articles and periodical literature follow a different form as stated earlier while explaining the entries in the bibliography.

Certain English and Latin abbreviations are quite often used in bibliographies and footnotes to eliminate tedious repetition. The following is a partial list of the most common abbreviations frequently used in report-writing (the researcher should learn to recognise them as well as he should learn to use them):

- non., anonymous
- ante., before
- art., article
- aug., augmented
- bk., book
- bull., bulletin
- cf., compare
- ch., chapter
- col., column
- diss., dissertation
- ed., editor, edition, edited.
- ed. cit., edition cited
- e.g., *exempli gratia*: for example
- eng., enlarged
- et.al., and others

et seq.,	et sequens: and the following
ex.,	example
f., ff.,	and the following
fig(s).,	figure(s)
fn.,	footnote
ibid., ibidem:	in the same place (when two or more successive footnotes refer to the same work, it is not necessary to repeat complete reference for the second footnote. Ibid. may be used. If different pages are referred to, pagination must be shown).
id., idem:	the same
ill., illus., or	
illust(s).	illustrated, illustration(s)
Intro., intro.,	introduction
<i>I</i> , or <i>II</i> ,	line(s)
loc. cit.,	in the place cited; used as op.cit., (when new reference
loco citato:	is made to the same pagination as cited in the previous note)
MS., MSS.,	Manuscript or Manuscripts
N.B., nota bene:	note well
n.d.,	no date
n.p.,	no place
no pub.,	no publisher
no(s).,	number(s)
o.p.,	out of print
op. cit:	in the work cited (If reference has been made to a work
opera citato	and new reference is to be made, ibid., may be used, if intervening reference has been made to different works, op.cit. must be used. The name of the author must precede.
p. or pp.,	page(s)
passim:	here and there
post:	after
rev.,	revised
tr., trans.,	translator, translated, translation
vid or vide:	see, refer to
viz.,	namely
vol. or vol(s).,	volume(s)
vs., versus:	against

8. Use of statistics, charts and graphs: A judicious use of statistics in research reports is often considered a virtue for it contributes a great deal towards the clarification and simplification of the material and research results. One may well remember that a good picture is often worth more than a thousand words. Statistics are usually presented in the form of tables, charts, bars and line-graphs and pictograms. Such presentation should be self explanatory and complete in itself. It should be suitable and appropriate looking to the problem at hand. Finally, statistical presentation should be neat and attractive.

9. The final draft: Revising and rewriting the rough draft of the report should be done with great care before writing the final draft. For the purpose, the researcher should put to himself questions like: Are the sentences written in the report clear? Are they grammatically correct? Do they say what is meant? Do the various points incorporated in the report fit together logically? “Having at least one colleague read the report just before the final revision is extremely helpful. Sentences that seem crystal-clear to the writer may prove quite confusing to other people; a connection that had seemed self evident may strike others as a *non-sequitur*. A friendly critic, by pointing out passages that seem unclear or illogical, and perhaps suggesting ways of remedying the difficulties, can be an invaluable aid in achieving the goal of adequate communication.”

LAYOUT OF THE RESEARCH REPORT

The layout of the report means as to what the research report should contain. A comprehensive layout of the research report should comprise

(A) preliminary pages; (B) the main text; and (C) the end matter

(A) Preliminary Pages

In its preliminary pages the report should carry a *title and date*, followed by acknowledgements in the form of 'Preface' or 'Foreword'. Then there should be a *table of contents* followed by *list of tables and illustrations* so that the decision-maker or anybody interested in reading the report can easily locate the required information in the report.

(B) Main Text

The main text provides the complete outline of the research report along with all details. Title of the research study is repeated at the top of the first page of the main text and then follows the other details on pages numbered consecutively, beginning with the second page. Each main section of the report should begin on a new page. The main text of the report should have the following sections: (i) Introduction; (ii) Statement of findings and recommendations; (iii) The results; (iv) The implications drawn from the results; and (v) The summary.

(i) *Introduction*: The purpose of introduction is to introduce the research project to the readers. It should contain a clear statement of the objectives of research i.e., enough background should be given to make clear to the reader why the problem was considered worth investigating. A brief summary of other relevant research may also be stated so that the present study can be seen in that context. The hypotheses of study, if any, and the definitions of the major concepts employed in the study should be explicitly stated in the introduction of the report.

(ii) *Statement of findings and recommendations*: After introduction, the research report must contain a statement of findings and recommendations in non-technical language so that it can be easily understood by all concerned. If the findings happen to be extensive, at this point they should be put in the summarised form.

(iii) **Results**: A detailed presentation of the findings of the study, with supporting data in the form of tables and charts together with a validation of results, is the next step in writing the main text of the report. This generally comprises the main body of the report, extending over several chapters. The result section of the report should contain statistical summaries and reductions of the data rather than the raw data. All the results should be presented in logical sequence and splitted into readily identifiable sections. All relevant results must find a place in the report.

(iv) **Summary**: It has become customary to conclude the research report with a very brief summary, resting in brief the research problem, the methodology, the major findings and the major conclusions drawn from the research results.

Contexts for writing – paper, thesis, reports etc.

What is clear to a recipient depends both on what is transmitted and how the recipient interprets it. Therefore, communicating clearly requires awareness of what the recipient brings. What is the recipient's background? What is the recipient seeking? How does the recipient expect the writing to be organized? Clarity in scientific writing requires attentiveness to such questions. As communication professionals advise, know your audience. Also know the conventions, and thus the expectations, for structuring the type of writing that you are doing.

a) Paper: A scientific paper is a written and published report describing original research results. A scientific paper must be written in a certain way, as defined by tradition, editorial practice, scientific ethics, and the interplay of printing and publishing procedures. Sufficient information must be presented so that potential users of the data can (1) assess observations, (2) repeat experiments, and (3) evaluate intellectual processes.

A scientific paper should be first providing broad orientation, then focusing narrowly on the specific research, and then considering the findings in wider context. Some have likened this shape for a scientific paper to an hourglass: broad, then narrow, then broad. Keeping this overall structure in mind can aid when writing individual parts of a paper and integrating them into a coherent whole.

b)Thesis :A Ph.D. thesis in the sciences is supposed to present the candidate's original research. Its purpose is to prove that the candidate can do and communicate such research. Therefore, a thesis should exhibit the same type of disciplined writing that is required in a journal publication. Unlike a scientific paper, a thesis may address more than one topic, and it may present more than one approach to some topics. The thesis may present all or most of the data obtained in the student's thesis-related research. Therefore, the thesis usually is longer and more involved than a scientific paper. But the concept that a thesis must be a bulky 200-page tome is wrong, dead wrong. Many 200-page theses contain only 50 pages of good science. The other 150 pages comprise turgid descriptions of insignificant details.

C)Reports :A scientific report is a document that describes the process, progress, and or results of technical or scientific research or the state of a technical or scientific research problem. It might also include recommendations and conclusion of the research

Tools for document preparation-LATEX

LaTeX is a powerful document markup language text processing system available for preparing impressive and highly structured documents. LaTeX is becoming the “Standard” text processor for publications in science, engineering, mathematics and report writing to scientific community. LaTeX is a family of programs integrates many tools needed to design, develop, documents in one application. It gives amazing results and particularly a deserving tool when working with mathematical symbols. This tool is an open source, where new features can be added to the software for further enhancement virtually by anyone who is interested in.

Latex is a document preparation system that internally uses tags to prepare a readable document with crystal clarity. Document preparation with Latex typically consists of using a text editor like Vi Editor, Notepad, Emacs to edit a Latex source file which has the extension as .tex file. Then, run the latex program to convert the source file to a document interchange format such as PostScript or PDF. Once the document is converted to PostScript or PDF format, it can be previewed on the screen, sent to others, and printed.

LaTeX has many features:

- Distinguished excellent print quality!
- Equations are relatively easy to enter, once you understand the programming aspects
- Latex automatically numbers the sections, sub- sections, figures, tables, foot notes, equations, theorems, lemmas, and the references of your paper. When you add a new section or sub-section, or a new reference, every item is re-numbered automatically. It can create its own table of contents and indices automatically as well.
- Spacing between Word is uniform throughout the document. In contrast, MS Word has this drawback when you justify a paragraph.
- It can easily manage very lengthy documents like books and theses.

Merits of LATEX Documentation

- Superior Typographical Quality: Latex facilitates unique typographical quality; however, one must spend time for familiarizing the tags and appropriate usage of each.
- Built-in Fonts: It supports all categories of fonts as available in MS-Word and Open office. In addition to that additional fonts like tiny, footnote size, normal size, large, huge are available.
- Good spacing algorithm •

Ability to fine-tune spacing arbitrarily: Latex is provided with additional tags for finetuning the space arbitrarily.

- Cross Platform/Portability: Latex can run virtually on any operating system in existence. MS-Word is compatible with windows and does not work on Unix/Linux/Mac os machines.
- Document Longevity: Latex documents written ten years ago still works and produce the same output as they did when originally written. In contrast, Microsoft uses version evolution as a strategic tool to force people to upgrade continuously. Consequently, MS Word documents are typically useful only for 3-4 years.

- **Mathematical Typesetting:** LaTeX includes all the capabilities of TeX in Mathematics typesetting and with few easier user interfaces. There are packages like amsmath which enhance and refine these interfaces. This tool facilitates all Greek and Hebrew letters, Relational Algebraic symbols, trigonometric symbols, binary operation, arrays, and vector symbols. Any complex equations can also be typed in this tool. LaTeX continues to excel in this task.
- **Cross-referencing capability:** LaTeX automatically assigns labels for tables, figures and equations while it is typed. You can refer to a table or equation with the corresponding label during the course of appending the document. If any mismatch in cross-referencing LaTeX compiler will report an error.
- **Macros and other programmatic features:** LaTeX allows us to define macros, which contain sequences of text and/or mark-up. It's equivalent to copy and paste in MS Word documents. Instead of copy and paste, here the same macro name is called in different locations of the document. LaTeX allows people to write programs in their documents. Moreover, any graphical diagrams can be drawn using a utility called tikz.

Demerits of Using Latex :

- Difficulty in Learning: User must be a computer literate to learn and make use of mark up language.
- Preview Delay: Every time there is time delay between the time tags are prepared and to see the result in PDF or PostScript.
- Possibility of Syntax errors: Unlike in a “What you see is what you get” [WYSIWYG] word processor, you need to create .tex file. Latex software will reject if there is violation in syntax. It takes time for learning the tags, debugging and compilation.
- Adding New font: You need to write code for adding new font tags in to the software.